Breaking Bouba-Kiki: The effect of label similarity on labelling perceptually similar referents

Introduction

Labels and the concepts they refer to are vital to language. When we encounter a novel label, what assumptions do we make, if any, of the thing it refers to? Although classical views emphasize the arbitrariness of linguistic form (de Saussure, 1916; Hockett, 1960), numerous studies have shown that people can in fact form intuitions based on the perceptual resemblance between a novel label and its potential referent (e.g., sound symbolism; Köhler, 1929; Ramachandran & Hubbard, 2001; for review, Dingemanse et al., 2016). Here we examine a complementary notion: that people have a bias to map similar labels onto similar concepts (and vice versa; see also Sloutsky & Fisher, 2012). Across three experiments in adults, we show bidirectional evidence of this bias and demonstrate that it can even reverse other baseline biases.

Experiment 1

In Experiment 1 (*N*=202), participants read short English sentences (*preambles*) featuring two novel labels (e.g. "There is a distant planet with two species called teki and boma."). On half the trials, before the preamble, we also introduced a "context" label and creature (e.g., "This is a toku. This toku lives on a distant planet […]"; Figure 1A, top). The context label (*toku*) was similar to one of the labels in the preamble (the *target label*; e.g. *teki*). Participants were then presented with one of the labels from the preamble and two new creatures (one of which, the *target creature*, looked similar to the context creature). Their task was to judge which creature best matched the label (e.g., "which of these is more similar to your idea of a teki?"; Figure 1A, bottom). We analyzed how often participants selected the target creature across context type (with/without a context creature) and label type (target/control label). We observed the expected interaction (B = 0.44, $p < .001$; Figure 2A): when cued with the target label, participants selected the target creature more often when they had previously seen a context creature with a similar name and appearance—a *label-similarity effect*.

Experiment 2

Experiment 2 (*N*=162) was designed similarly to Experiment 1, with two main differences: we added catch trials to exclude inattentive participants, and we created and pre-tested stimuli to have baseline sound symbolic label-image biases. This allowed us to test whether the label-similarity effect is capable of overriding other non-arbitrary preferences, rather than merely reflecting them. As intended, these stimuli were clearly non-arbitrary at baseline (Figure 2B, left side). Critically, we successfully reversed these baseline preferences with context: there was a significant interaction between context type and label type (B = 0.48, $p < .001$; Figure 2B), replicating Experiment 1 and showing that in this paradigm, the similarity-based bias can reverse pre-existing non-arbitrary biases.

Experiment 3

In Experiment 3 (*N*=78), participants *freely generated their own labels* for creatures, avoiding the limitations of a forced choice task. As before, they read short preambles with or without a context creature, but then produced (by typing out) their own labels for another creature (Figure 1B). We analyzed the Levenshtein similarity between these responses and the context label as a function of context type and label type. Our results again showed the expected interaction: freely generated labels for the target creature were more similar to the context label when a similar context creature had been introduced (B = 0.33, $p < .001$; Figure 2C).

Summary

We demonstrate a *label-similarity effect* – people prefer to map similar labels onto perceptually similar referents. Because such mappings are not predicted by some principles of linguistic efficiency (Gibson et al., 2019 for review), these findings present a puzzle and suggest that understanding this bias may shed light on how linguistic and conceptual structures interact in the mind.

Figure 1: Schematic showing experiment design

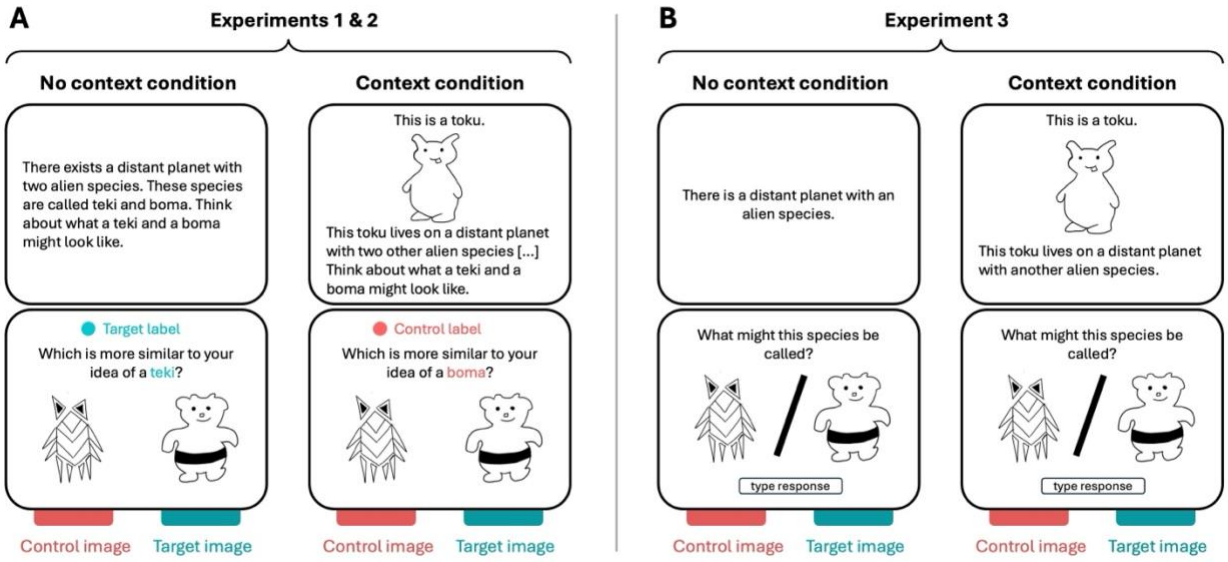
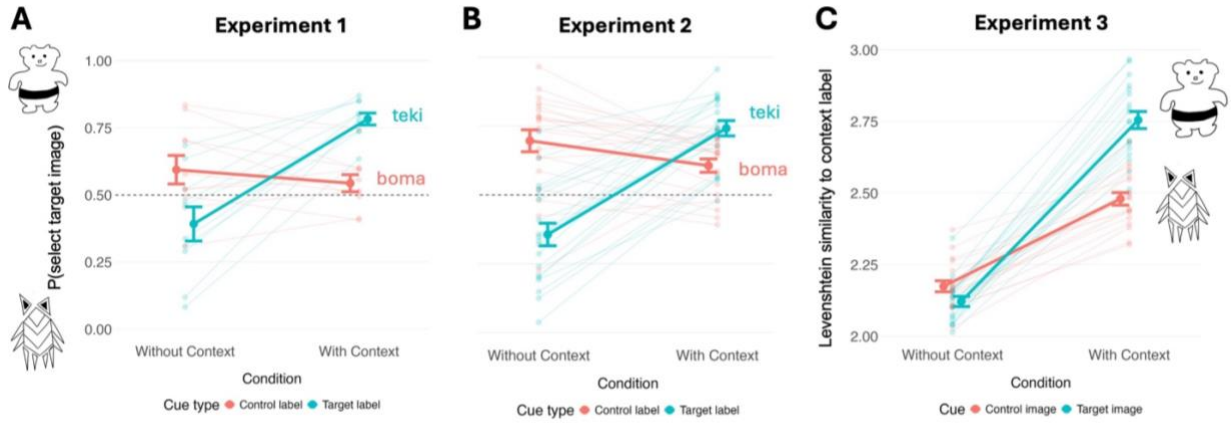


Figure 2: Experiment results



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